

Requirements Analysis:

Project Name

Human Development Index (HDI) Predictor

Introduction:

The Human Development Index (HDI) Predictor is a Machine Learning-based web application that predicts the Human Development Index of a country using Life Expectancy, Education Index, and Income Index. This document describes the functional and non-functional requirements of the project.

Functional Requirements

- The application should allow users to enter Life Expectancy.
- The application should allow users to enter Education Index.
- The application should allow users to enter Income Index.
- The application should validate user inputs.
- The application should predict the Human Development Index (HDI).
- The application should display the predicted HDI.
- The application should provide a simple and user-friendly interface.
- Non-Functional Requirements
- The application should provide accurate predictions.
- The system should respond quickly.
- The application should be reliable.
- The user interface should be simple and easy to use.
- The application should be available through a web browser.

Software Requirements

- Python
- Flask
- Pandas
- NumPy
- Scikit-learn
- HTML
- CSS
- GitHub
- VS Code

Hardware Requirements

- Computer/Laptop or Mobile
- Internet Connection
- Minimum 4 GB RAM
- Modern Web Browser

Users

- Students
- Researchers
- Government Organizations
- Data Analysts
- General Users

Expected Outcome

The Human Development Index (HDI) Predictor should accurately predict the HDI using Life Expectancy, Education Index, and Income Index. The application should provide fast, reliable, and user-friendly predictions for users.